

ARAVINDSELVAN C

Generative AI Engineer | AI/ML Developer

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PROFESSIONAL SUMMARY

Final-year B.E. ECE student at Sri Sairam Engineering College with hands-on expertise in machine learning, generative AI, and full-stack development. Engineered 25+ end-to-end AI/ML systems spanning RL agents, latent diffusion pipelines, and NLP engines. Secured IEEE funding of \$5,420 for a medical AI prototype (AirTon) and reached the finals of the IMC Prosperity 4 global algorithmic trading competition. Certified by Google, AWS, IIT Bombay, and NPTEL.

EDUCATION

B.E. – Electronics & Communications Engineering | Sri Sairam Engineering College, Chennai

Sep 2023 – May 2027

CGPA: 6.95 / 10.0

Class 12 – Computer Mathematics | Akshaya Academy Matric Hr. Sec. School, Dindigul

June 2022 – May 2023

Score: 86% / 100%

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, TypeScript, Java

AI / ML: Machine Learning, Generative AI, fine-tuning, RAG, TensorFlow, Scikit-learn, Pandas, NumPy, Hugging Face Transformers, Gemini API, Reinforcement Learning, NLP, LLMs, Vector Databases, Prompt Engineering

Web & APIs: React.js, Node.js, FastAPI, HTML5, CSS3, REST APIs, TypeScript

Cloud & DevOps: Google Cloud Platform, AWS, Vercel, Render.com

Hardware & IoT: Arduino, ESP32, Firmware Prototyping, Real-time Sensor Interfacing, Embedded Systems

Tools: Claude Code, VS Code, Jupyter Notebook, Cursor AI, GitHub Copilot, Blockchain, GitHub, Linux

INTERNSHIP EXPERIENCE

Machine Learning Intern — Prodigy InfoTech (Remote) | Ref: CIN PIT/NOV23/3787

Nov – Dec 2023

- **Developed** and evaluated supervised regression models for housing price prediction; received outstanding performance remarks.
- **Built** end-to-end Python ML pipelines — data preprocessing, feature engineering, model training, hyperparameter tuning — across 5+ algorithms.

PROJECTS

AirTon — Handheld Medical AI Device for screening and detection of glaucoma

2024 – 2027

Python · TensorFlow · ESP32 · Computer Vision · IoT

- **Engineered** an ML-powered hardware prototype for non-invasive glaucoma screening via IOP detection; awarded IEEE funding of \$5,420.
- **Integrated** real-time sensor data pipelines with custom ESP32 firmware for live biometric capture and on-device classification.

Tradenza — Reinforcement Learning Trading Agent (XAUUSD) 2026

Python · TensorFlow · Reinforcement Learning · Pandas · NumPy

- **Architected** an autonomous RL trading agent for XAUUSD with custom reward functions incorporating Sharpe Ratio and max-drawdown metrics.
- **Implemented** a full backtesting pipeline to evaluate strategy performance; agent generates buy/sell/hold signals with zero manual intervention.

Ghibli Art Generator — Stylized Latent Diffusion Pipeline

2025

Python · PyTorch · Stable Diffusion SDXL/SD 1.5 · LoRA · Hugging Face Diffusers · Gradio

- **Engineered** a text-to-image & image-to-image pipeline with SDXL + LoRA fine-tuning for high-fidelity Studio Ghibli style transfer.
- **Deployed** public demo on Hugging Face Spaces via Gradio with zero infrastructure overhead.

edumate — Smart Student Dashboard

2026

TypeScript · React.js · Node.js · FastAPI · Python · Vercel · Render

- **Designed** and deployed a mobile-optimized student dashboard for Sri Sairam Engineering College with real-time attendance and data visualization.
- **Achieved** <2 s load times across Vercel (frontend) + Render (Python FastAPI backend) deployment stack.

Resume-Architect — AI-Powered Resume Optimization Engine

2026

Python · Gemini API · LLMs · NLP · Vector Databases · Google AI Studio

- **Developed** an end-to-end NLP pipeline that parses, scores, and benchmarks resumes against job descriptions using LLM embeddings and vector similarity deployed via Google AI Studio .

Health Monitoring System & Blockchain Visualization Tool

2024

JavaScript · Node.js · REST APIs · HTML5 Canvas

- **Built** a real-time health monitoring web app with biometric alert thresholds; created an interactive blockchain visualizer using HTML5 Canvas animations.

ACHIEVEMENTS & COMPETITIONS

- IMC Prosperity 4 — Finalist, Global Algorithmic Trading Competition
- AirTon Medical Device — Awarded \$5,420 IEEE funding for AI-based glaucoma detection prototype
- MSME Hackathon 2024 — Top 50 teams nationally
- Smart India Hackathon 2025 — Wait-listed
- Google Solution Challenge — Participant, GDSC | Flipkart GRiD 6.0 — Level 1 cleared | Adobe India Hackathon — Round 1 qualified
- Google Tunix Hackathon — AI Lead, Model Fine-Tuning | Global Virtual Summit 2024 — Delegate, GELP

CERTIFICATIONS

- Google Cloud / Coursera — Introduction to Generative AI; AI Skills Challenge Level 3: GenAIus Registries
- Amazon Web — Introduction to Machine Learning on AWS
- IIT Bombay Spoken Tutorial — Advanced C++ (Score: 85%), Java (Score: 67.5%)
- NPTEL — Data Science with Python (62%); Computer Networks and Internet Protocol (62%)
- HackerRank — Software Engineer Certificate; Python Basic & Problem Solving Basic
- LinkedIn Learning — Advanced Prompt Engineering Techniques | GUVI / Skill India (NSDC) — AI For India 2.0
- Anthropic / Claude — Claude Code 101; AI Fluency for Students | Google / Coursera — Foundations: Data, Data, Everywhere

COMPETITIVE PROGRAMMING & ONLINE PROFILES

LeetCode: leetcode.com/Aravindselvan | **HackerRank:** hackerrank.com/aravindselvan201 | **SkillRack:** skillrack.com/profile/441693
Microsoft Learn: learn.microsoft.com/users/aravindselvanc-2555 | **GeeksForGeeks:** geeksforgeeks.org/user/aravindselvan2006